

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (EC) Examination

Nov-Dec 2012

EC 202 Network Theory

Date: 28.11.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

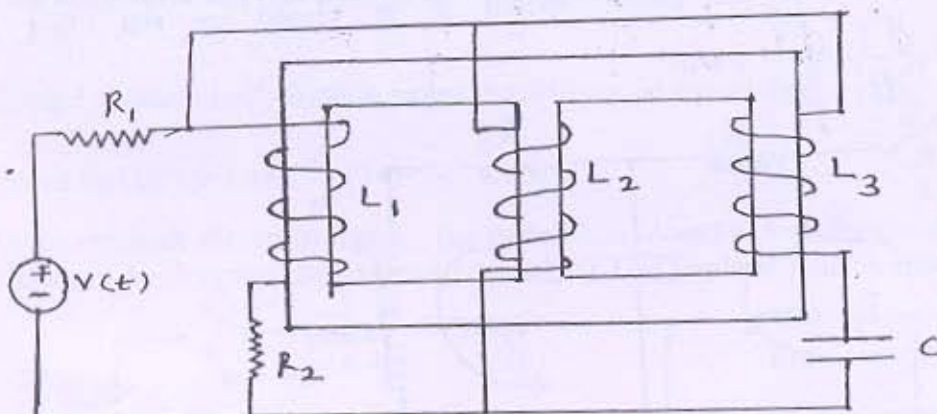
Maximum Marks: 70

Instructions:

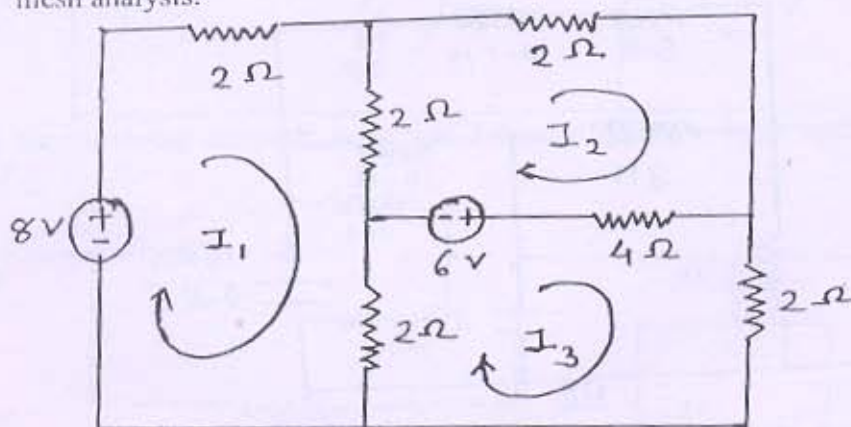
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) (1) Explain KVL and KCL with suitable Example. [02]
(2) Classification of Networks. [03]
(3) Explain Source Transformation Rules (any five). [02]
- Q - 2 (a) The following Network shows three windings on a magnetic core. Using different shaped dots establish polarity markings for the windings and using specified currents write the Kirchhoff's voltage law equations for this network. [07]

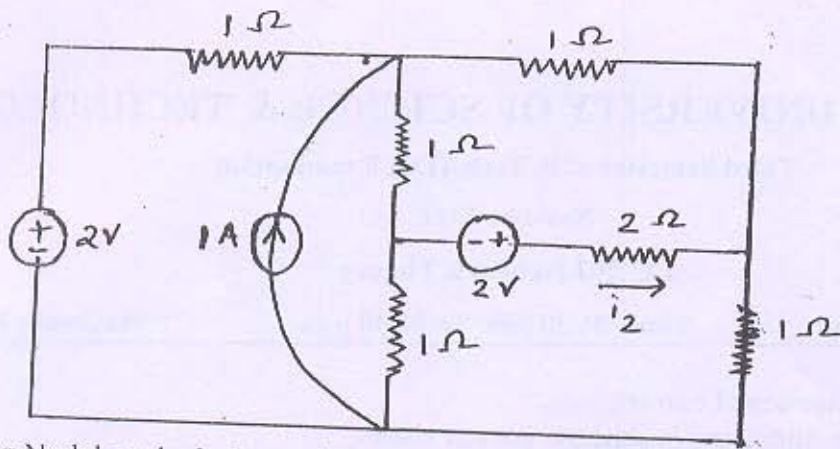


- (b) Determine the current through the 4Ω resistor branch of the given network using mesh analysis. [07]



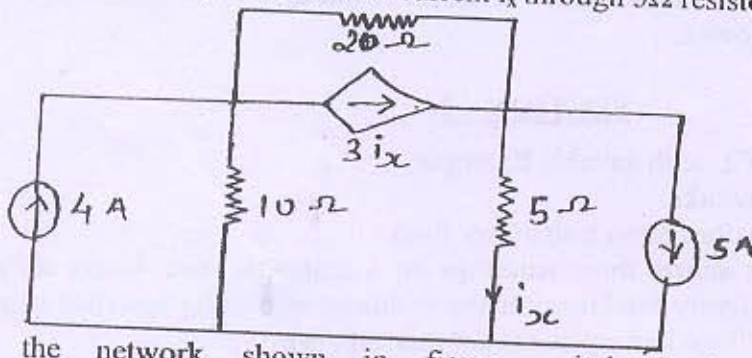
OR

- Q - 2 (a) In the following network determine the value of i_2 using source transformation method. [07]



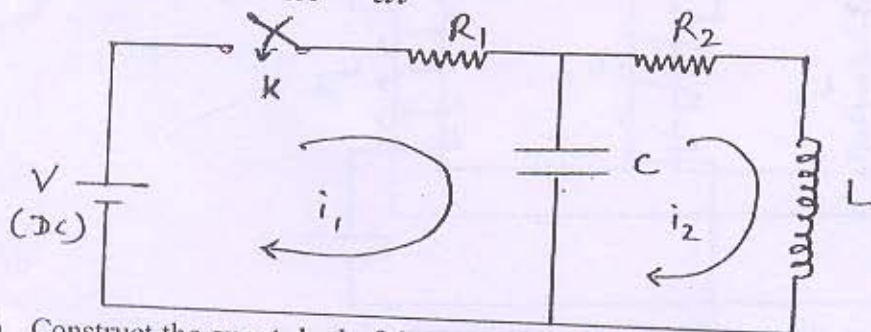
(b) Using Nodal analysis determine current i_x through 5Ω resistor.

[07]



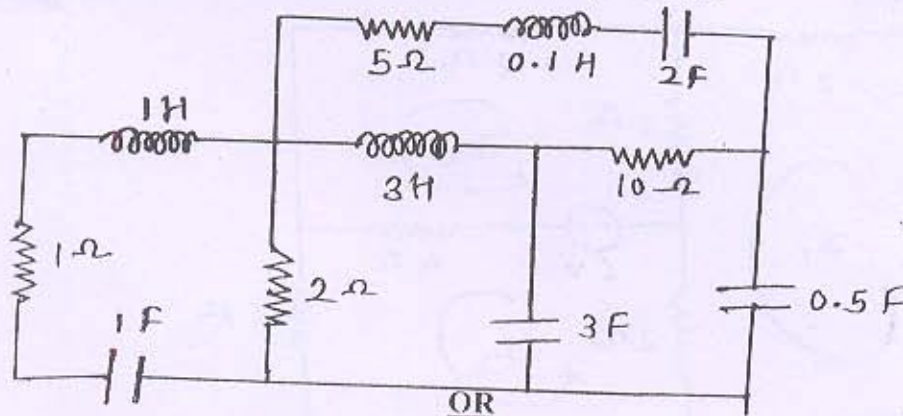
Q-3 (a) In the network shown in figure switch k is closed at $t=0$, find [09]

$$i_1, i_2, \frac{di_1}{dt}, \frac{di_2}{dt}, \frac{d^2i_1}{dt^2}, \frac{d^2i_2}{dt^2} \text{ at } t=0+.$$



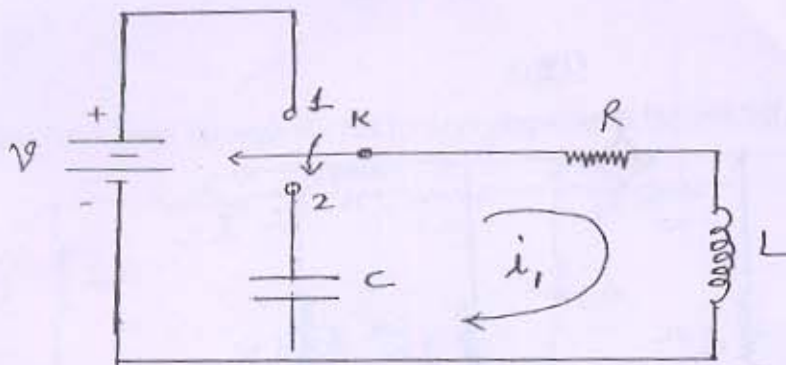
(b) Construct the exact dual of the network of following network.

[05]

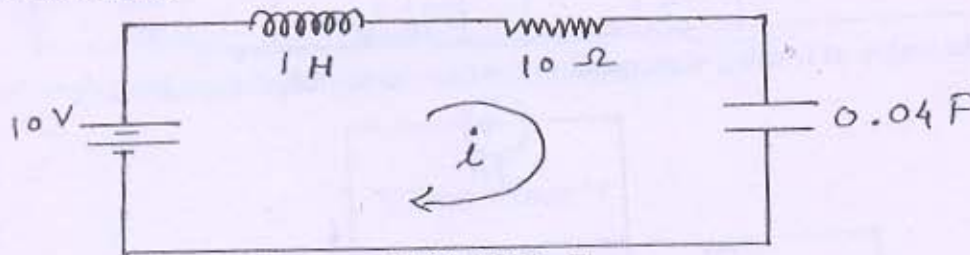


OR

Q-3 (a) In the following network, the switch k is changed from position 1 to 2 at $t=0$. Find [07]
values of i_1 , $\frac{di_1}{dt}$ and $\frac{d^2i_1}{dt^2}$ at $t=0+$, if $R=1000\Omega$, $L=1H$, $C=0.1\mu f$ and $V=100$ volts.



- (b) In the following network a d.c. voltage of 10 volts is suddenly applied to a series RLC circuit. The capacitor is initially uncharged. Obtain the particular solution for current $i(t)$ in the circuit. [07]



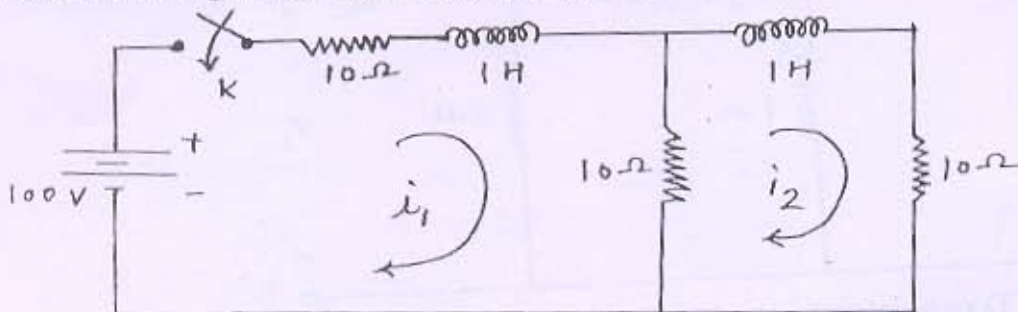
SECTION - II

- Q - 4 (a) Find the inverse Laplace transform of the function $\frac{s-1}{s^2+3s+2}$. [03]

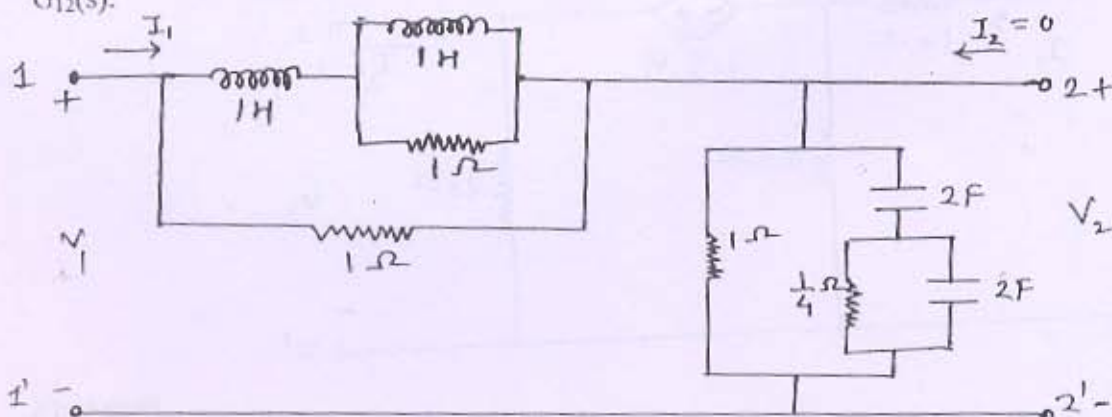
- (b) Using Laplace transformation, solve the differential equation $\frac{d^2i}{dt^2} + 4\frac{di}{dt} + 8i = 8u(t)$. [04]

Given that $i(0^+) = 3$ and $\frac{di}{dt}(0^+) = -4$.

- Q - 5 (a) In the network shown in figure, the switch k is closed at $t=0$. for the element values shown in the diagram, determine $i_1(t)$ and $i_2(t)$. Use Laplace Transformation. [07]



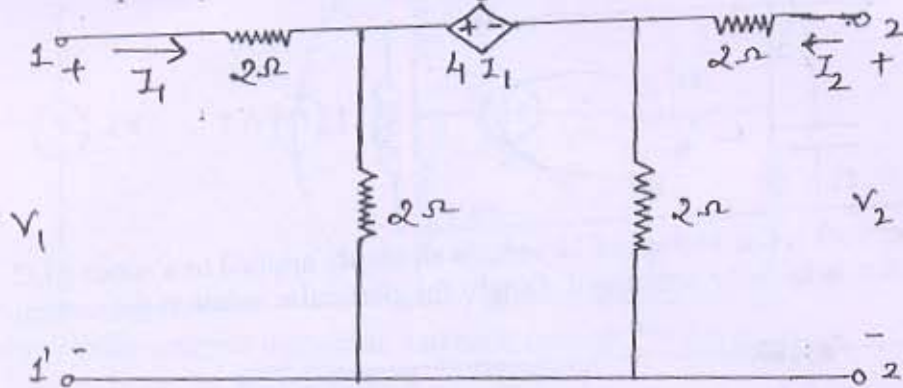
- (b) For the following network, with port 2 open, find the voltage ratio transfer function $G_{12}(s)$. [07]



OR

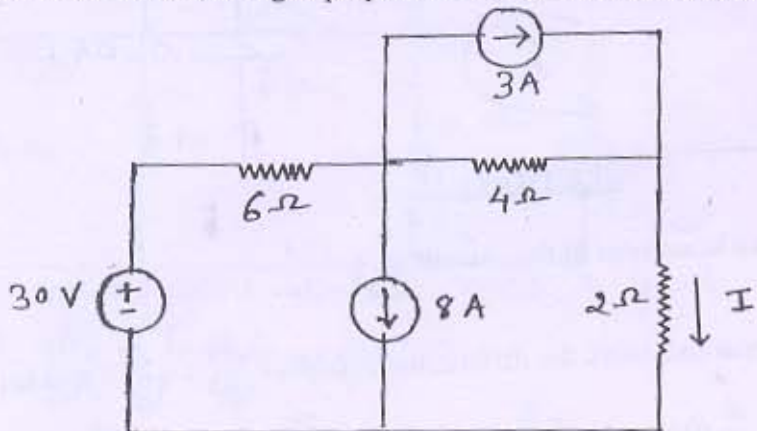
Q - 5 (a) Find the Z-parameters for the following network.

[07]



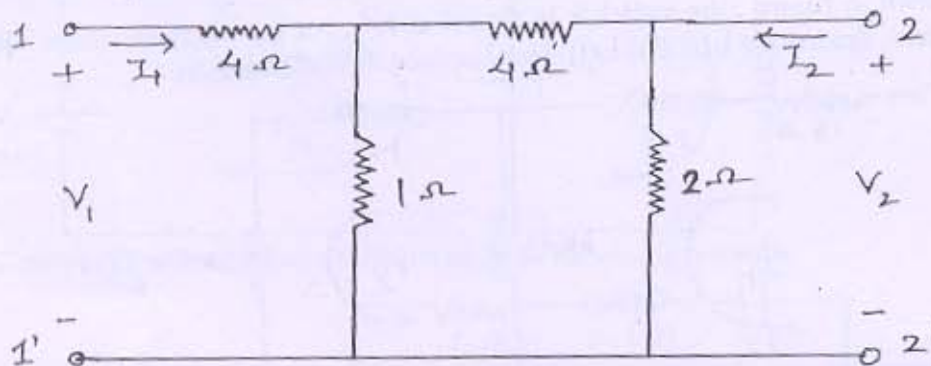
(b) Determine the value of i_1 using superposition theorem in the following network.

[07]



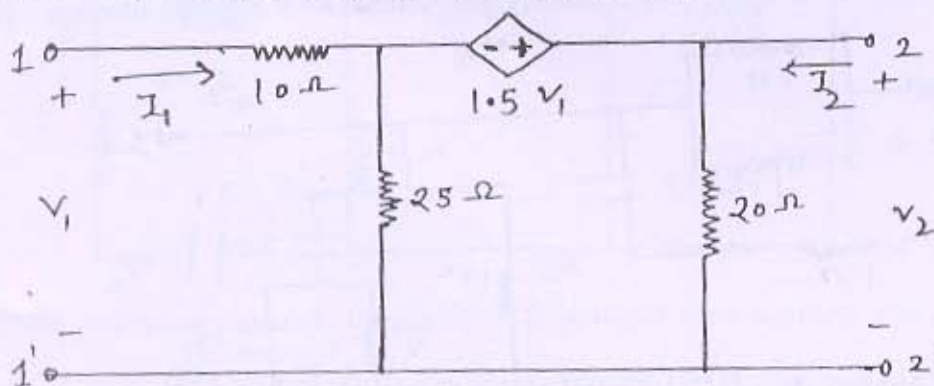
Q - 6 (a) Find the h-parameters for the following network.

[07]



(b) Find the ABCD parameters for the following network.

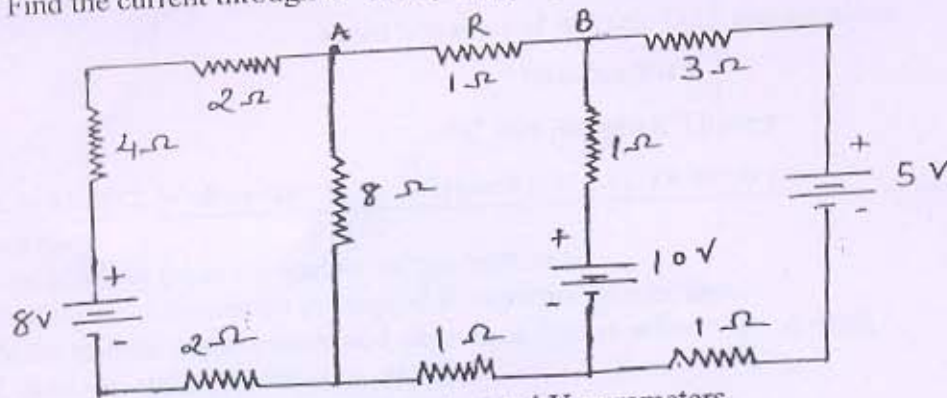
[07]



OR

[07]

Q - 6 (a) Find the current through $R=1\Omega$ by applying Thevenin's Theorem.



[07]

(b) Relationship between Z-parameters and Y-parameters.

Best of Luck